# DA2400+DA2401 — Machine Learning I (+ lab)

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|---|---|
| Tier | A |
| Phase (order) | 3 |
| Do this first | DA1001, DA1004, DA1300, DA1002 |
| Estimated effort | 14 weeks, ~10 h/week |
| Unlocks | DA3407, DA2403, DA3409, DA5400 |

## Roadmap

1. Regression, bias-variance, Lasso/Ridge.
2. Classification: Naive Bayes, logistic regression, SVM, trees, boosting, random forests.
3. Unsupervised: PCA, k-means.
4. Lab: pandas, missing data, model comparison.

## Video courses (all links checked on YouTube on 2026-10-22)

### 🎓 Main course: follow it start to end

- Machine Learning Foundations (93 videos). IIT Madras BS *Machine Learning Foundations*: the IITM DS-programme ML start.

### 🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Machine Learning Techniques ( Prof. Arunkumar ) (74 videos). IIT Madras BS *Machine Learning Techniques* (Arun Rajkumar): PCA, clustering, regression, SVM, ensembles.
- NPTEL MOOC Machine Learning 2016 (88 videos). NPTEL *Introduction to Machine Learning* (Balaraman Ravindran, IIT Madras).
- Stanford CS229: Machine Learning led by Andrew Ng | Autumn 2018 (21 videos). Stanford CS229 (Andrew Ng, Autumn 2018, official): the maths-heavy classic.
- Machine Learning Practice Course (98 videos). IIT Madras BS *Machine Learning Practice*: scikit-learn, for the lab (DA2401).

### ⚡ Short videos for intuition and revision

- Machine Learning (100+ videos). StatQuest *Machine Learning*: one clear video per algorithm.

## Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

| Week | Lectures | What to do |
|---|---|---|
| 1 | 1–9 | What ML is: regression, classification, dimensionality reduction, density estimation. |
| 2 | 10–16 | Univariate and multivariate calculus, linear approximation. |
| 3 | 17–24 | Four subspaces, orthogonality, projections, least squares. Implement least squares regression in NumPy. |
| 4 | 25–31 | Polynomial regression, eigenvalues, diagonalisation. |
| 5 | 32–37 | Complex, Hermitian and unitary matrices. |
| 6 | 38–44 | SVD, positive definite matrices. |
| 7 | 45–49, 93 | PCA. Implement PCA from scratch and apply it to MNIST. |
| 8 | 50–57 | Unconstrained optimisation, gradient descent, Taylor series. |
| 9-10 | 58–71, 92 | Constrained optimisation, Lagrange multipliers, convexity, duality, KKT. |
| 11 | 72–86 | Probability: discrete and continuous random variables, expectation, common distributions. |
| 12 | 87–91 | Multivariate normal, parameter estimation, GMM and EM, concentration inequalities. Implement EM for a 1-D GMM. |

▼ **All lectures in *Machine Learning Foundations* (93 videos)**

1. W1.L1. What is machine learning? | intro to machine learning & its applications
2. W1_L2: Data, models & machine learning task
3. W1_L3: Supervised learning: regression
4. W1_L4: Supervised learning: classification
5. W1_L5: Unsupervised learning: dimensionality reduction
6. W1_L6: Unsupervised learning: density estimation
7. W1_T1: To machine learning or not to machine learning
8. W1_T2: Illustration with a real world dataset
9. W1_T3: Dimensionality reduction & density estimation with applications
10. W2_L1: Sets & functions
11. W2_L2: Univariate calculus: continuity & differentiability
12. W2_L3: Univariate calculus: derivatives & linear approximations
13. W2_L4: Univariate calculus: applications & advanced rules
14. W2_L5: Multivariate calculus: lines & planes in higher dimensional space
15. W2_L6: Multivariate calculus: linear approximation & applications
16. Week 2: Tutorial
17. W3_L1: Four fundamental subspaces
18. W3_L2: Orthogonal vectors & subspaces

19. W3_L3: Projections
20. W3_L4: Least squares & projections onto a subspace
21. W3_L5: Example of least squares
22. W3_T1: Four fundamental vector subspaces
23. W3_T2: Row space computation using reduced row echelon form & solution to ax = b
24. W3_T3: Orthogonality, projections & least squares method
25. W4_L1: Linear & polynomial regression
26. W4_L2: Eigenvalues & Eigenvectors
27. W4_L3: Diagonalization of a matrix
28. W4_L4: Solving fibonacci sequence using diagonalization
29. W4_L5: Orthogonality diagonalizable matrices
30. W4_T1: Polynomial regression
31. W4_T2: Tutorial on eigenvalues & eigenvectors
32. W5_L1: Complex matrices
33. W5_L2: Hermitian matrices
34. W5_L3: Unitary matrices
35. W5_L4: Diagonalization of hermitian matrices | part 1
36. W5_L5: Diagonalization of hermitian matrices | part 2
37. W5_T1: Some problems to think & solve
38. W6_L1: Singular value decomposition
39. W6_L2: Example of singular value decomposition (SVD)
40. W6_L3: Positive definiteness
41. W6_L4: Positive definite matrices
42. W6_T1: Real world application of SVD
43. W6_T2: Geometric interpretation of SVD
44. W6_T3: Positive definite functions & matrices
45. W7_L1: Principal component analysis(PCA)
46. W7_L2: Principal component analysis (contd.)
47. W7_L3: PCA as maximizing variance
48. W7_L4: PCA in higher dimensions
49. W7_T1: Tutorial | principal component analysis
50. W8_L1: Pillars of machine learning
51. W8_L2: Introduction to optimization
52. W8_L3: Solving an unconstrained optimization problem | part 1
53. W8_L4: Solving an unconstrained optimization problem | part 2
54. W8_L5: Basic algorithm for unconstrained optimization: gradient descent
55. W8_L6: Gradient descent & taylor series
56. W8_L7: Gradient descent for multivariate functions
57. W8_L8: Taylor series in higher dimensions
58. W9_L1: Constrained optimization | part -1
59. W9_L2: Constrained optimization | part -2

60. W9_L3: Method of lagrange multiplier, projected gradient descent
61. W9_L4: Introduction to convexity
62. W9_L5: Properties of convex sets
63. W9_L6: Convex functions
64. W9_L7: Properties of convex functions
65. W9_T1: Constrained optimization, lagrange multpliers, convex sets & convex functions
66. W10_L1: Properties of convex functions
67. W10_L2: Applications of optimization in machine learning
68. W10_L3: Revisiting constrained optimization
69. W10_L4: Relation between primal & dual problem, karush-kuhn-tucker(KKT) conditions
70. W10_L5: Karush-kuhn-tucker (KKT) conditions continued
71. W10_T1: Linear programming, KKT conditions, relationship between primal & dual problem
72. Probability space
73. Conditioning
74. Random variables
75. Expectation:definition & properties
76. Multiple random variables
77. Bernoulli, binomial, poisson & geometric random variables
78. Solved examples
79. W11_L1: Continuous random variables
80. W11_L2: Conditional pdf
81. W11_L3: Expectation
82. W11_L4: Multiple random variables
83. W11_L5: Independent random variables
84. W11_L6: Transformed random variables
85. W11_L7: Uniform, exponential, normal
86. W11_T1: Examples on continuous random variables
87. W12_L1: Bivariate & multivariate normal
88. W12_L2: Estimation of parameters using machine learning
89. W12_L3: Gaussian mixture models & expectation maximization
90. W12_L4: Laws of large numbers: markov, chebyshev, hoeffding, central limit
91. W12_T1: Solved examples
92. Revisiting Constrained Optimization
93. Week 7: Tutorial

## Books, notes and websites

- ✅ NPTEL *Introduction to Machine Learning* (Ravindran, IIT Madras) — nptel.ac.in/courses/106106139/
- Stanford CS229; Book: *ISLP* (statlearning.com, free) — the lab text
- Book: Bishop PRML; Deisenroth *Mathematics for ML* (mml-book.com)
- scikit-learn user guide; Kaggle Learn

## Practice / proof of understanding

Implement 5 algorithms from scratch in NumPy; finish 3 Kaggle tabular competitions with proper CV.

## What your IITM course sheet says

### From the PDF: DA2400

**Official description:** The students should be able to 1. Identify the type of machine learning problem suited to the task at hand, based on the problem characteristics and data availability. 2. Design the details of the ML problem like evaluation metrics, data preprocessing 3. Construct basic algorithms for the most common type of ML problems. 4. Implement these algorithms on the computer for a variety of settings.

**Official syllabus (trimmed):**

> 1. Foundations. (1 week) Linear algebra, probability, optimisation, data, models, learning algorithms, train and test data.
> 2. Regression. (3 weeks) Squared loss minimisation, regularisation, closed form and optimisation based solutions: pseudo-inverse and (mini-batch, stochastic) gradient descent, overfitting/underfitting, bias-variance decomposition, Lasso regression.
> 3. Classification (3 weeks) Generative and discriminative models for classification, naive Bayes classifiers, logistic regression, multiclass reductions, discriminant methods, softmax classifiers, support vector machines, k-nearest neighbour, decision trees, ensemble methods: bagging, boosting, random forests.
> 4. Unsupervised Learning (3 weeks) Dimensionality reduction: principal component analysis, encoders and de- coders, reconstruction error. Clustering: k-means, Lloyd’s algorithm, hierarchical clustering, inertia, Rand-index. Density estimation: maximum likelihood, generative models, scoring and sampling based methods. Frequent pattern mining: basket analysis, bump finding, apriori algorithm.
> 5. Advanced Topics (3 weeks) Neural networks and automatic differentiation. Other paradigms of learning: rein- forcement learning, self-supervised learning, semi-supervised learning, active learning, online learning. Other ML tasks and evaluation metrics: ranking, structured prediction, confusion matrix,

**Official textbooks:** 1. Marc Peter Deisenroth, A. Aldo Faisal and Cheng Soon Ong. Mathematics for Machine Learning. Cambridge University Press. 2. Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani and Jonathan Taylor. An Introduction to statistical learning: with applications in Python. 3. Christopher Bishop. Pattern Recognition and Machine Learning. Springer.

**Official references:** 1. Marc Peter Deisenroth, A. Aldo Faisal and Cheng Soon Ong. Mathematics for Machine Learning. Cambridge University Press. 2. Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani and Jonathan Taylor. An Introduction to statistical learning: with applications in Python. 3. Christopher Bishop. Pattern Recognition and Machine Learning. Springer.

### From the PDF: DA2401

**Official description:** The students should be able to 1. Construct basic algorithms for the most common type of ML problems. 2. Implement these algorithms on the computer for a variety of settings. 3. Identify failure modes of various algorithms and debug them.

**Official syllabus (trimmed):**

> 1. Foundations. (1 week) Linear equation solving, simulations and histograms, gradient descent, python tools for machine learning.
> 2. Regression. (3 weeks) Model selection, analysis of bias and variance, Lasso regression.
> 3. Classification (4 weeks) Naive Bayes classifier, logistic regression, support vector machines, decision trees, boost- ing, random forests.
> 4. Unsupervised Learning (2 weeks) PCA for compression, clustering using Lloyd's algorithm,
> 5. Other topics (4 weeks) Handling multiple data files, data preprocessing, handling missing data, comparing performance of various algorithms, statistical tests of significance. Automatic differentiation with Pytorch. Learning neural network models.

**Official textbooks:** 1. Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani and Jonathan Taylor. An Introduction to statistical learning: with applications in Python. 2. Marc Peter Deisenroth, A. Aldo Faisal and Cheng Soon Ong. An Introduction to statistical learning: with applica- tions in Python. Cambridge University Press. 3. Christopher Bishop. Pattern Recognition and Machine Learning. Springer.

**Official references:** 1. Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani and Jonathan Taylor. An Introduction to statistical learning: with applications in Python. 2. Marc Peter Deisenroth, A. Aldo Faisal and Cheng Soon Ong. An Introduction to statistical learning: with applica- tions in Python. Cambridge University Press. 3. Christopher Bishop. Pattern Recognition and Machine Learning. Springer.

## From the PDF: DA5400

**Official description:** The objective of this course is to introduce fundamentals of machine learning techniques and their applications in different problems.

**Official syllabus (trimmed):**

> a. Representation Learning - PCA b. Estimation - Review of MLE, Bayesian estimation c. Clustering - K-Means, Hierarchical Clustering, Spectral Clustering 2. Supervised Learning: a. Regression b. Linear Regression, Ordinary Least Squares, PCR c. Non-linear regression (basis functions) d. Ridge Regression, LASSO 3. Binary Classification: a. K-Nearest Neighbors b. Decision Trees, CART c. Bias-Variance Dichotomy, Model Validation: Cross validation d. Bayesian Decision Theory e. Generative vs Discriminative Modeling for classification i. Generative 1. Naive Bayes, Gaussian Discriminant Analysis 2. Hidden Markov Model ii. Discriminative 1. Logistic Regression 4. Advanced Methods for Classification a. Support Vector Machines - Kernels b. Ensemble Methods: i. Bagging - Random Forest ii. Boosting - Adaboost/GBDT/XgBoost iii. Artificial Neural Networks iv. Multi-Class Classification - one vs all, one vs one 5. Sequential Decision Making

**Official textbooks:** 1. Hastie, T., Tibshirani, R., Friedman, J. H., & Friedman, J. H. (2009). The elements of statistical learning: data mining, inference, and prediction. New York: Springer.2. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2013). An introduction to statistical learning. New York: Springer.

**Official references:** 1. Duda, R. O. & Hart, P. E. (2006). Pattern Classification. John Wiley & Sons.2. Bishop C. M. & Nasrabadi. N. M. (2006). Pattern Recognition and Machine Learning, New York: Springer.

## From the PDF: DA5401

**Official description:** Covers basics of python or R. Simple analytics tasks - regression, classification,clustering, associations, etc. Emphasis will be on choice of models, evaluation of results, significance analysis, visualization and interpretation of results.

**Official syllabus (trimmed):**

> 1.Introduction to various Python toolkits: Numpy for handling arrays and matrices; SciPy for scientific computing; Matplotlib for data visualization; Pandas for data manipulation; SciKit Learn library for machine learning.2.Linear models for regression: Ordinary least squares; Ridge regression (l2 regularization); Lasso (l1 regularization); Elastic Net (l2-l1 regularization).3.Linear classification: Linear Discriminant Analysis (LDA); Logistics regression; Linear Support Vector Machine (SVM); l2 and l1 regularized versions of these algorithms.4.Non-linear algorithms: Kernel SVM, Random forrest. Neural network.5.Unsupervised learning: Dimensionality reduction technique such as Principal Component Analysis (PCA), Clustering techniques such as k-Means clustering and Agglomerative clustering

**Official textbooks:** 1.Sarah Guido, Andreas C. Müller, Introduction to Machine Learning with Python, O'Reilly Media, Inc., 2016.2.Jerome H. Friedman, Robert Tibshirani, and Trevor Hastie, The Elements of Statistical Learning, Second Edition (Springer Series in Statistics)

**Official references:** 1.Edouard Duchesnay, Tommy Löfstedt, Statistics and Machine Learning in Python, Draft by the authors, available online.